

Supplement S2: Finite validation evidence

These computations are independent finite validation evidence, not premises of the all- k proofs. The semantic reference implementation enumerates strict labelled total orders and checks directed M/V precedence and same-target Butterfly noncrossing exactly.

The bounded replay covers all 64 period words at $k = 2$, exact canonical $MMMMVV$ and $MVMMMM$ pivot fibres, all five exact symmetry actions at $k = 2$, and all 64 formal words at $k = 1$ grouped into 16 physical projections.

The observed $k = 2$ distribution is 48×0 , 8×1 , 4×7 , 4×5 , for 56 labelled states total. Exact symmetry set equality passes 320/320 checks. At $k = 1$ there are 32 singleton and 32 empty formal words, invariant under changes of the absent u_1, l_1 bits.